anti-gravity-sb-landowner-sync Project — Technical Specification

Document type: Technical Specification | Project: anti-gravity-sb-landowner-sync | Version: 1.0.0 | Date: June 2026 | Platform: MuleSoft Anypoint Platform 4.6.0 | **Proposed Design** | Source: Land_Owners_Requirements.pdf

1. Project Overview

This integration project syncs land-rights owner records from the Rainbow system (polled via OData REST API) to the Salesforce custom object `LandOwner__c` as flat records. It runs on a nightly scheduler. All artifacts, flows, and deployments are prefixed with `anti-gravity-` to distinguish this implementation.

Metadata Property	Value
Maven Group ID	`08756342ee7847158fd70786d2c83b4e`
Maven Artifact ID	`anti-gravity-sb-landowner-sync`
Version	`1.0.0`
Packaging	`mule-application`
Mule Runtime	4.6.0 **Proposed**
Mule Maven Plugin	4.1.0 **Proposed**
HTTP Port	`8081` (Operational/Health only)

Flow definition file(s)	`anti-gravity-global-config.xml`
	`anti-gravity-error-handling.xml`
	`anti-gravity-landowner-sync.xml`

2. Connector Dependencies

Connector	Group ID	Version	Purpose
`mule-http-connector`	`org.mule.connectors`	1.10.6 **Proposed**	Inbound HTTP listener & outbound HTTP requests to OData API

Connector	Group ID	Version	Purpose
`mule-sockets-connector`	`org.mule.connectors`	1.2.5 Proposed	Low-level socket support (runtime dependency)
`mule-salesforce-connector`	`com.mulesoft.connectors`	11.4.0 Proposed	Salesforce database upsert and deletion operations
`mule-secure-configuration-property-module`	`com.mulesoft.modules`	1.2.7 Proposed	Secure decryption of system properties

3. Global Configurations

3.1 HTTP Listener Configuration — `anti-gravity-http-listener-config`

Property	Value / Reference
Host	`0.0.0.0`
Port	`${http.port}` (default 8081)
doc:id	`c389478f-64d8-4ba8-9b88-51f7871b6910`

3.2 HTTP Request Configuration (Rainbow Analytics API) — `anti-gravity-rainbow-analytics-request-config`

Property	Value / Reference
Protocol	`HTTPS`
Host	`${rainbow.analytics.host}`
Port	`443`
Connection Timeout	`${rainbow.timeout.ms}` (30000ms)
Authentication	Basic Authentication (Username: `${rainbow.username}`, Password: `${secure::rainbow.password}`)
doc:id	`b37c093c-28df-419b-a3d8-55452d0a1b9c`

3.3 HTTP Request Configuration (Rainbow Operation API) — `anti-gravity-rainbow-operation-request-config`

Property	Value / Reference
Protocol	`HTTPS`
Host	`${rainbow.operation.host}`
Port	`443`
Connection Timeout	`${rainbow.timeout.ms}` (30000ms)
Authentication	Basic Authentication (Username: `${rainbow.username}`, Password: `${secure::rainbow.password}`)
doc:id	`a5ef4811-92df-422f-8da7-111122223333`

3.4 Salesforce Configuration — `anti-gravity-salesforce-config`

Property	Value / Reference
Auth Type	Basic (Username/Password)
Username	`${salesforce.username}`
Password	`${secure::salesforce.password}`
Security Token	`${secure::salesforce.securityToken}`
doc:id	`d7cf12a4-44df-41ef-ba38-1234567890ab`

3.5 Configuration Properties — File Declarations

File Path	Description
`config.yaml`	Non-secure configurations (Hosts, Ports, Cron schedules)
`secure-config.yaml`	Secure encrypted credentials, decrypted using system key `secure.key`

4. Flows

4.1 Flow: `anti-gravity-landowner-sync-scheduler-flow`

doc:id: 887ac0c2-55ef-4560-8438-fa12c3b4d5e1

Scheduler Trigger:

Trigger	Expression / Interval	Description
Scheduler	`${sync.cron}` (default `0 0 2 * * ?`)	Runs nightly at 02:00 AM Asia/Jerusalem timezone

Processing Steps:

1. **Log START** — Logs `anti-gravity-sb-landowner-sync - anti-gravity-landowner-sync-scheduler-flow - START`.
2. **Call Delete Sub-flow** — Executes `anti-gravity-delete-stale-landowners-subflow` to remove outdated records before downloading new records.
3. **Call Fetch Registrations Sub-flow** — Executes `anti-gravity-fetch-registrations-subflow` to retrieve all current properties.
4. **ForEach Registration** — Loops over each retrieved active property registration.
    - Calls `anti-gravity-build-landowners-for-registration-subflow` to fetch owners and map them.
5. **Choice: Empties Guard** — Evaluates if landowners list is empty.
    - If NOT empty: Calls `anti-gravity-upsert-landowners-subflow`.
    - If empty: Logs a warning and skips upsert.
6. **Log END** — Logs `anti-gravity-sb-landowner-sync - anti-gravity-landowner-sync-scheduler-flow - END`.

4.2 Sub-flow: `anti-gravity-delete-stale-landowners-subflow`

doc:id: b3df18aa-f17a-42a9-9134-8c88722ba3e1

Description: Finds and deletes Salesforce landowner records whose `ToDate__c` is in the past, except those manually assigned to a lawyer.

Processing Steps:

1. **Salesforce Query** — Run query: `SELECT Id FROM LandOwner__c WHERE ToDate__c < TODAY AND Lawyer__c = null`.
2. **Choice: Empty Check** — If records are found:
    - Deletes the records from Salesforce by ID in chunks.

4.3 Sub-flow: `anti-gravity-fetch-registrations-subflow`

doc:id: 9d784a9e-f823-44ca-8ba2-c2834bfa12de

Description: Queries the OData endpoint to fetch current active registrations, supporting next-link pagination.

Processing Steps:

1. **HTTP GET Registrations** — Calls `GET /registrations` with query parameters: `$filter=isCurrentRegistration eq true`.
2. **Loop / Paging Process** — Check for `@odata.nextLink`. Recursively pulls next pages if available and concatenates items.
3. **Transform Data** — Maps registrations payload using `anti-gravity-map-registrations.dwl` into a list of simplified registration objects.

4.4 Sub-flow: `anti-gravity-build-landowners-for-registration-subflow`

doc:id: 8bfa9321-df10-41ab-8e01-12c83ba01de3

Description: Fetches owner properties and their associated business partner cards, building flat landowner models.

Processing Steps:

1. **HTTP GET Owners** — Calls `GET /owners` with query parameters: `$filter=toDate eq null and registrationKey eq [registration.key]`.
2. **Choice: Has Owners?** — If owners are found:
    - Loops through each owner.
    - Calls `GET /businessPartners` with query parameters: `$filter=id eq [owner.businessPartnerId]`.
    - Maps owner, registration, and partner details into the flat Salesforce structure using `anti-gravity-build-landowner.dwl`.
    - Executes `anti-gravity-add-building-lookup.dwl` to attach building reference if `buildingCode` is present.
    - Accumulates the results in a session variable.

4.5 Sub-flow: `anti-gravity-upsert-landowners-subflow`

doc:id: f092df44-12af-4dfd-9b1b-aa11b22cc33d

Description: Performs bulk upserts of accumulated landowner data to Salesforce.

Processing Steps:

1. **Batch Chunking** — Splits the landowner list into chunks based on `${salesforce.batchSize}` (default 200).
2. **Salesforce Upsert** — Calls Upsert on `LandOwner__c` using external key `External_Key__c`.
3. **Process Results** — Loops through upsert results. Logs errors for any single item failure, reporting the associated `External_Key__c`.

4.6 Flow: `anti-gravity-health-flow`

doc:id: a23fa990-21af-42e3-8da9-a2a3bcbfdecc

HTTP Trigger:

Method	Path	Port	Description
GET	`/health`	`${http.port}` (8081)	Basic application health status check

Processing Steps:

1. **Set Payload** — Sets response body to `"OK"`.

Response:

HTTP Status	Content-Type	Expected Response Body
200	`text/plain`	`OK`

5. Configuration Properties

Property Key	Default / Placeholder	Description
`http.port`	`8081`	Operational endpoints API HTTP port
`sync.cron`	`0 0 2 * * ?`	Cron expression for nightly sync triggers
`rainbow.analytics.host`	`ziv-rainbow.net`	Rainbow OData Analytics endpoint host name
`rainbow.operation.host`	`ziv-rainbow.net`	Rainbow OData Operational endpoint host name
`rainbow.timeout.ms`	`30000`	Rainbow HTTP request connection and read timeout limit
`rainbow.username`	`CHANGE_ME`	Username for OData API access *(replace before running)*
`rainbow.password`	`CHANGE_ME`	Secure password for OData API access *(replace before running)*
`salesforce.username`	`CHANGE_ME`	Salesforce connection username *(replace before running)*
`salesforce.password`	`CHANGE_ME`	Secure Salesforce connection password *(replace before running)*

Property Key	Default / Placeholder	Description
`salesforce.securityToken`	`CHANGE_ME`	Secure Salesforce API access token *(replace before running)*
`salesforce.apiVersion`	`59.0`	Target Salesforce SOAP/REST API Version
`salesforce.batchSize`	`200`	Batch size chunk limit for Salesforce Bulk/Soap Upsert

6. Error Handling Strategy

Error Type	HTTP Status	Response Format / Behavior
`HTTP:CONNECTIVITY`, `HTTP:TIMEOUT` (Rainbow)	N/A	On Error Continue. Retry 3 times with backoff, log failed registration, continue processing remaining registrations.
`SALESFORCE:CONNECTIVITY`	N/A	On Error Propagate. Terminate scheduler flow execution, log critical ERROR, alert system admins.
`SALESFORCE:INVALID_FIELD`, record errors	N/A	On Error Continue (upsert). Log failing record `External_Key__c`, proceed with rest of the batch.
`ANY`	500 (HTTP) / N/A (Scheduler)	Global default handler: Log full error track with transaction correlation ID.

7. Logging Convention

Event	Level	Message Pattern
Flow Start	INFO	`anti-gravity-sb-landowner-sync - [Flow Name] - START`
OData Pull Success	INFO	`anti-gravity-sb-landowner-sync - Fetch registrations - Count: [records count]`
Salesforce Upsert Fail	ERROR	`anti-gravity-sb-landowner-sync - Salesforce Upsert failed for External Key: [key]`
Flow End	INFO	`anti-gravity-sb-landowner-sync - [Flow Name] - END`

Logging uses a rolling file appender target writing to `logs/anti-gravity-sb-landowner-sync.log`, configured to limit files to 10 MB with 10 max backup archives.

8. External API Reference

8.1 Rainbow OData API

Base URL	Method & Path	Key Parameters	Usage
`https://ziv-rainbow.net/api/analytics/odata`	`GET /registrations`	`$filter=isCurrentRegistration eq true`	Fetch properties containing rights details
`https://ziv-rainbow.net/api/analytics/odata`	`GET /owners`	`$filter=toDate eq null and registrationKey eq [key]`	Fetch owners of specific properties
`https://ziv-rainbow.net/api/operation/odata`	`GET /businessPartners`	`$filter=id eq [id]`	Retrieve legal identity card fields for partners

9. Data Transformation Detail

Purpose: Maps registrations, owners, and business partner cards into flat Salesforce records.

DataWeave (build-landowner):

```
%dw 2.0
output application/java
---
payload map (item) -> {
    Name: item.businessPartnerName[0..79],
    Idtype__c: item.idType,
    IdNumber__C: item.legalIdNumber,
    Address__c: item.address,
    Essence__c: item.rightType,
    OwnershipPercent__c: item.ownershipPercent,
    External_Key__c: item.ownerId as String,
    Project__r: {
        Project_Code__c: item.projectCode
    },
    (if (item.buildingCode != null) {
        Building__r: {
            BuildingCode__c: item.buildingCode
        }
    } else {}),
```

```
    Plot__c: item.plot,
    ToDate__c: item.toDate
}
```

10. Build & Run

```
# Build
mvn clean package

# Build without tests
mvn clean package -DskipTests

# Deploy to local Mule runtime
mvn clean package -DmuleDeploy
```

11. Sample Request & Response

11.1 GET /health

Request:

```
GET http://localhost:8081/health
```

Response Status: `200 OK`
Response Body:

```
OK
```

12. Design Assumptions & Open Questions

Design Assumptions:

Assumption	Rationale	Impact if wrong
Mule Runtime 4.6.0	Modern, long-term supported Java 17 release	Compatibility issues with older runtimes if user runs 4.4.x
Salesforce API v59.0	Stable version that supports upserting related external ID Lookups natively	Need to rewrite relationship maps if version is too old

Assumption	Rationale	Impact if wrong
Basic Authentication for Rainbow API	Direct credentials indicated in connection properties	Auth failures if Rainbow enforces OAuth or token exchanges

Open Questions:

Question	Why it matters	Suggested Default
Rainbow OData API Rate Limiting	Paging registrations recursively could hit threshold caps	Assume no strict API rate limits for nightly sync runs
Building code lookup behavior	Confirm if building code values always match Salesforce master Building table exactly	Assume codes match, otherwise upserts fail with lookup exceptions

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